

AXIOM PROTOCOL

Security Model

Threat model, wallet safety, AI agent protection, upgrade safety, and mainnet readiness checklist.

Trust Mathematics, Not Power.

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Security goals

- Protect monetary immutability: no unauthorized inflation, emergency mint, or hidden supply path.
- Protect ownership: no protocol-level confiscation, freeze, rollback, or privileged seizure.
- Protect consensus: prevent double-signing, invalid finality, validator cartel capture, and liveness failure.
- Protect users: reduce phishing, malicious approvals, wallet loss, and contract interaction risk.
- Protect AI agents: prevent unlimited spending, runaway transaction spam, and agent hijacking.
- Protect decentralization: keep node operation affordable and client diversity healthy.

Threat model

Axiom should assume rational attackers, wealthy attackers, malicious validators, bot farms, compromised mobile devices, phishing websites, malicious smart contracts, dishonest AI agents, spam attacks, MEV extraction, governance capture, bridge attacks, and regulatory pressure. The system should reduce damage even when one layer fails.

Wallet security model

Axiom should use smart accounts as the default user experience. Users should not be forced to manage raw seed phrases. Security should be built into the account design.

Wallet controls

- Passkey or hardware-backed device key for everyday signing.
- Social recovery or guardian recovery with timelock.
- Spending limits and allowlists for new users.
- Transaction simulation before signing.
- Contract risk warnings and verified-app badges.
- Emergency lock for the user account only, not protocol-level freezing.
- Session keys for games, AI agents, and micropayments.

AI agent security

AI safety rule: an AI agent may spend only what it is allowed to spend, where it is allowed to spend, for as long as it is allowed to spend, with full revocation by the owner.

The main wallet should never be exposed directly to the model, plugin, tool, or agent runtime. AI actions should be logged and explainable to the owner.

Consensus security

Use proven PoS+BFT in early testnets before adding experimental reputation or mobile participation. Use validator caps, square-root effective stake, randomized committees, slashing, public performance dashboards, client diversity, and adversarial testnets.

MEV and fee security

If Axiom supports DeFi, it must handle MEV risk. Possible tools include encrypted mempools, fair ordering, batch auctions, MEV burn or redistribution, and user protection modes. Fees should be adaptive to block-space demand to reduce spam and reflect market conditions.

Upgrade security

Axiom must be upgradeable enough to fix bugs and improve cryptography, but not upgradeable enough to become centrally controlled money. Upgrades should use public proposals, audit windows, testnet deployment, timelocks, multi-client release, and voluntary node adoption. Monetary rules remain outside governance.

Mainnet readiness checklist

- Publish formal specification and reference implementation.
- Run public testnets with adversarial incentives.
- Perform independent audits for consensus, VM, wallet, vesting contracts, and fee market.
- Run economic simulations for validator sustainability and Sybil resistance.
- Launch bug bounty before mainnet candidate.
- Ensure reproducible builds and a client diversity plan.
- Remove all admin keys and prove no privileged mint, freeze, or pause paths exist.
- Publish genesis allocation and founder vesting contract address.
- Publish legal risk analysis for applications while keeping the core protocol neutral.
- Run final community review of the Immutable Charter.

Disclaimer

This document is a Phase 0 research and design specification. It is not investment advice, legal advice, security assurance, or a promise that the system is feasible without further research, simulations, audits, and testnet validation.